Direct Methods - NLP Problems

Introduction to Optimal Control Problem

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NLP Problems

NLP Problems

To solve optimization problems like those seen in class, it is necessary to use a numerical approximation (except for the simplest cases).

Direct methods transform:

Original problem:		Transformed problem:
Dynamic optimization	$\xrightarrow{transf.}$	**Nonlinear optimization**
Infinite dimension		**Finite dimension**

To perform this transformation, one must discretize the state and control variables correctly.

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Discretization

The time interval $[t_I, t_F]$ is divided into N subintervals. The resulting points are:

$$\{\bar{t}_0, \bar{t}_1, \ldots, \bar{t}_N\}$$

with $\bar{t}_0 := t_I$ and $\bar{t}_N := t_F$.

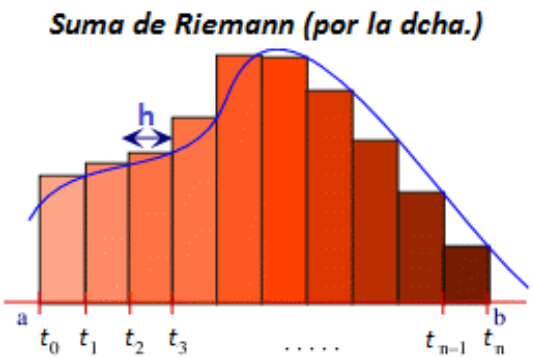
In each subinterval $[\bar{t}_i, \bar{t}_{i+1}]$, $i = 0, \ldots, N-1$, a numerical integration scheme is used for the differential equation:

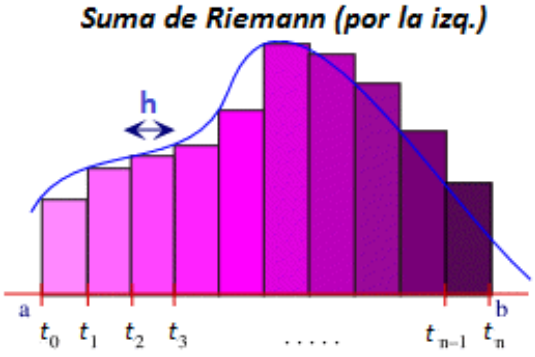
$$\mathbf{x}'(t) = \mathbf{g}[\mathbf{x}(t), u(t)].$$

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Integration

Basic methods. **Rectangle rule**:





$$\mathbf{x}(\bar{t}_{i+1}) - \mathbf{x}(\bar{t}_i) = \int_{\bar{t}_i}^{\bar{t}_{i+1}} \mathbf{g}[\mathbf{x}(t), u(t)]dt \approx h_i\, \mathbf{g}[\mathbf{x}(\bar{t}_i), u(\bar{t}_i)],$$

where $h_i = (\bar{t}_{i+1} - \bar{t}_i)$

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Integration

Basic methods. **Trapezoidal rule**:

$$\mathbf{x}(\bar{t}_{i+1}) - \mathbf{x}(\bar{t}_i) = \int_{\bar{t}_i}^{\bar{t}_{i+1}} \mathbf{g}[\mathbf{x}(t), u(t)]dt \approx$$

$$\approx \frac{h_i}{2}\left(\mathbf{g}[\mathbf{x}(\bar{t}_i), u(\bar{t}_i)] + \mathbf{g}[\mathbf{x}(\bar{t}_{i+1}), u(\bar{t}_{i+1})]\right),$$

where $h_i = (\bar{t}_{i+1} - \bar{t}_i)$

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Integration

Either rule can be used to approximate the solution of the differential equation in each subinterval, of the form:

$$\mathbf{x}'(t) = \mathbf{g}[\mathbf{x}(t), u(t)]$$

which can be expressed as "constraints."

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Integration

Denoting:

$$\mathbf{x}_i = \mathbf{x}(\bar{t}_i), \quad \mathbf{x}_{i+1} = \mathbf{x}(\bar{t}_{i+1})$$
$$u_i = u(\bar{t}_i), \quad u_{i+1} = u(\bar{t}_{i+1})$$

From the Rectangle Rule we obtain:

$$\mathbf{x}_{i+1} = \mathbf{x}_i + h_i \mathbf{g}(\mathbf{x}_i, u_i),$$

which, written in its most general form, becomes:

$$\mathbf{x}_{i+1} = \tilde{\mathbf{g}}(\mathbf{x}_i, u_i).$$

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Integration

Similarly, from the Trapezoidal Rule we get:

$$\mathbf{x}_{i+1} = \mathbf{x}_i + \frac{h_i}{2}\left[\mathbf{g}(\mathbf{x}_i, u_i) + \mathbf{g}(\mathbf{x}_{i+1}, u_{i+1})\right],$$

In general form:

$$\mathbf{x}_{i+1} = \tilde{\mathbf{g}}(\mathbf{x}_i, u_i, \mathbf{x}_{i+1}, u_{i+1}).$$

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Hermite–Simpson Method

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HS Method

Review

Different integration rules have already been introduced that can be used to approximate the solution of the differential equation in each subinterval, in the form:

$$\dot{y}(t) = f\left(y(t), u(t), t\right)$$

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HS Method

Integration

The **Rectangle Rule**, the simplest of this type, consists of taking the interpolating function to be constant (a zero–order polynomial).

We also studied the **Trapezoidal Rule**, which consists of approximating the function by a polygonal line (a first–order polynomial) whose segments are determined by two consecutive points taken on the curve representing the given function.

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HS Method

Simpson's Rule for Numerical Integration

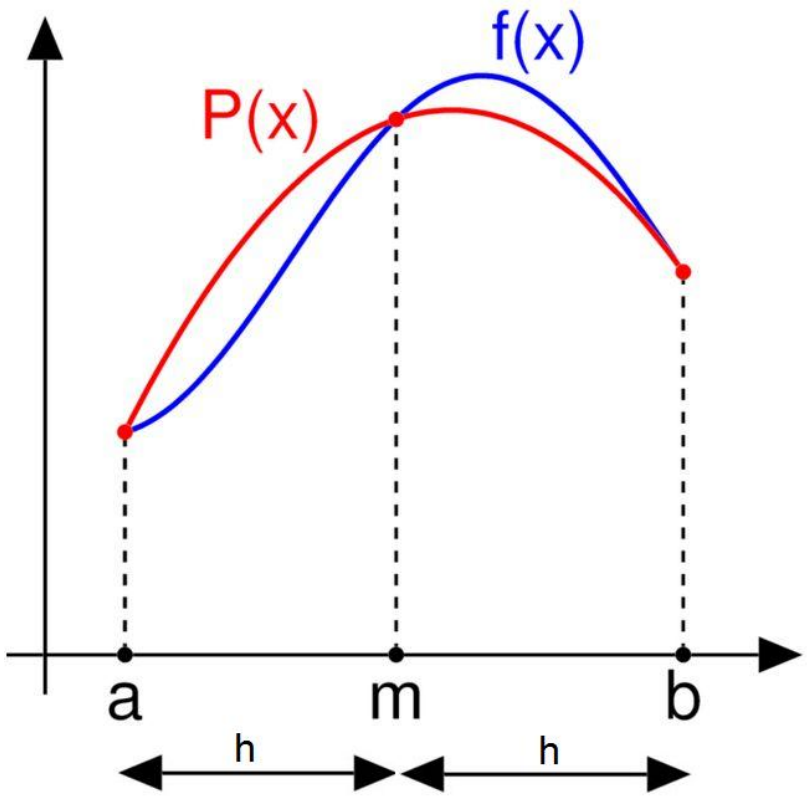
If instead of approximating the given function by a polygonal, we approximate it using **parabolic arcs** (second–order polynomials), the approximation improves and, consequently, the approximation of the definite integral of the given function becomes more accurate.

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HS Method

Simpson's Rule for Numerical Integration

SIMPLE SIMPSON'S RULE



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Hermite–Simpson Integration Method

For most applications, we need a more accurate method than the trapezoidal rule. The Hermite–Simpson rule is presented below:

Given the differential constraints:

$$\dot{y}(t) = f\left(y(t), u(t), t\right)$$

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Hermite–Simpson Integration Method

For this method, in any subinterval k, $k+1$:

$$0 = y_{k+1} - y_k - \frac{h_k}{6}\left[f_{k+1} + 4\bar{f}_k + f_k\right]$$

Where:

- y_k and y_{k+1} are the variable y evaluated at each end of the subinterval.
- f_k and f_{k+1} are the function f evaluated at each end of the subinterval.
- $\bar{f}_k$ is the function f evaluated at the midpoint of the subinterval k, $k+1$, that is:

$$\bar{f}_k = f\left(\bar{y}_k, \bar{u}_k, t_k + \frac{h_k}{2}\right)$$

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Hermite–Simpson Integration Method

To determine $\bar{f}_k = f\left(\bar{y}_k, \bar{u}_k, t_k + \frac{h_k}{2}\right)$, the following approximations are made:

- **State variables at the midpoint**, $\bar{y}_k$: are approximated using Hermite interpolation:

$$\bar{y}_k = \frac{1}{2}\left(y_{k+1} + y_k\right) + \frac{h_k}{8}\left(f_k - f_{k+1}\right)$$

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Hermite–Simpson Integration Method

To determine $\bar{f}_k = f\left(\bar{y}_k, \bar{u}_k, t_k + \frac{h_k}{2}\right)$, the following approximations are made:

- **Control variables at the midpoint**, $\bar{u}_k$: these variables are not approximated as the states. They are defined as new variables of the problem, with the same domain and minimum and maximum limits as the original control variable and can be approximated or not.

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Hermite–Simpson Integration Method

To determine $\bar{f}_k = f\left(\bar{y}_k, \bar{u}_k, t_k + \frac{h_k}{2}\right)$, the following approximations are made:

- Other **auxiliary variables** in the problem are approximated as the mean value of the original variables at the ends of the subinterval.